

Customer Segmentation & Churn Pattern Analytics in European Banking

Abstract

Customer churn is a major challenge in retail banking because customer attrition directly affects revenue, profitability, and long-term growth. This project analyzes churn behavior using a European banking customer dataset. The study applies data cleaning, exploratory data analysis, customer segmentation, KPI analysis, and visualization techniques to identify high-risk customer groups and uncover actionable business insights.

Introduction

Banks invest significant resources in acquiring customers, making retention a critical business objective. Understanding which customers are likely to leave and why they leave enables organizations to design targeted retention strategies. This project investigates churn patterns across geography, demographics, engagement, and financial characteristics.

Problem Statement

Despite having rich customer-level data, banks often lack detailed segmentation insights. The objective is to identify high-risk customer segments, evaluate geographic and demographic churn differences, and quantify financial exposure associated with churn.

Dataset Description

The dataset contains customer information including CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. The Exited column represents customer churn and serves as the primary analysis target.

Methodology

The analysis follows a structured workflow: data cleaning and validation, feature grouping and segmentation, exploratory data analysis, KPI calculation, churn distribution analysis, financial risk assessment, and business recommendation generation.

Customer Segmentation

Customers are segmented by geography (France, Germany, Spain), age groups, tenure groups, balance categories, and credit score bands. Segment-level churn rates are calculated to determine which groups contribute most to customer attrition.

Key Performance Indicators (KPIs)

KPIs include Overall Churn Rate, Segment Churn Rate, Geographic Risk Index, High-Value Customer Churn Ratio, Revenue at Risk, and Engagement Drop Indicator. These metrics provide a business-focused view of customer retention performance.

Exploratory Data Analysis

Visualizations include churn distribution, geography-wise churn, age-group churn, gender comparison, active versus inactive customer analysis, product usage analysis, balance distribution, credit score distribution, and correlation analysis. These charts reveal patterns and relationships

associated with churn.

Key Insights

Typical findings include elevated churn among inactive customers, geographic differences in churn rates, varying churn behavior across age groups, and potential revenue loss from high-balance customers. These insights help prioritize retention efforts.

Business Recommendations

Recommended actions include targeted retention campaigns for high-risk segments, improved engagement initiatives for inactive customers, loyalty programs for premium customers, and region-specific customer retention strategies.

Conclusion

The project demonstrates how customer segmentation and churn analytics can support data-driven decision making in banking. By identifying vulnerable customer groups and quantifying churn risk, organizations can improve retention outcomes and reduce revenue loss.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Streamlit, and GitHub were used to perform analysis, visualization, reporting, and deployment.